Sample Project Documentation

OptiCrop — Smart Agricultural Production Optimization Engine

1. Problem Statement

Farmers often select crops based on tradition or informal advice rather than measured soil and climate data, leading to lower yields and avoidable financial loss. OptiCrop replaces this guesswork with a trained ML model that recommends the most suitable crop for a given set of soil/climate conditions.

2. Dataset

Crop Recommendation Dataset (Kaggle) — 2,200 samples, 22 balanced crop classes (100 samples each), 7 features: N, P, K, temperature, humidity, pH, rainfall.

3. Methodology

- Exploratory Data Analysis: univariate (distributions), bivariate (humidity vs. crop), multivariate (correlation heatmap, pairplot).
- Pre-processing: null check (none found), IQR-based outlier handling on Potassium with log transform, derived seasonal feature, stratified 80/20 train/test split.
- Model building: K-Means clustering for exploratory grouping; Logistic Regression, KNN, Decision Tree, and Random Forest trained and compared as classifiers.
- Model selection: Random Forest selected based on highest Accuracy (99.55%), Precision, Recall, and F1-Score on held-out test data.
- Deployment: trained model serialized with pickle and served through a Flask web application with 3 pages (Home, About, Find Your Crop).

4. Results

Random Forest achieved 99.55% test accuracy, outperforming KNN (97.95%), Decision Tree (97.95%), and Logistic Regression (97.27%). The confusion matrix showed a strong diagonal with minimal misclassification, concentrated among agronomically similar crops.

5. Application

A 3-page Flask web application allows any user to enter soil/climate values and receive an instant crop recommendation, with an About page explaining the underlying methodology for transparency.

6. Repository

Full source code, dataset, trained model artifacts, and the training notebook are available at: https://github.com/neelimavana/OptiCrop

7. Conclusion

OptiCrop demonstrates a complete, reproducible ML lifecycle — from raw agricultural data through EDA, pre-processing, model comparison, and deployment — packaged into an accessible tool for farmers, researchers, and policymakers alike.